

Fundamentals of Solid State Physics

Thermal Properties

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Thermal Properties

- Heat Capacity (Thermal Capacity) 热容
- Thermal Expansion 热膨胀
- Thermal Conductivity 热导
- ...

Thermal Properties

- Thermal properties are the combinations of properties of **lattice vibration (phonons)** and **free electrons**
- For insulators, there are no free electron. Thermal properties of **lattice vibration (phonons)** dominate.
- For metals,
thermal properties = **phonon part + free electron part**

Thermal capacity

$$C_V = C_{V,p} + C_{V,e}$$

Thermal conductivity

$$\kappa = \kappa_p + \kappa_e$$

Fundamentals of Solid State Physics

Thermal Properties - Phonons

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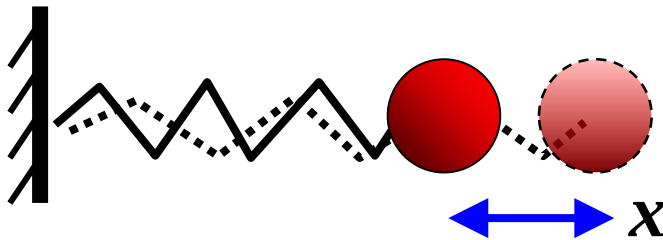
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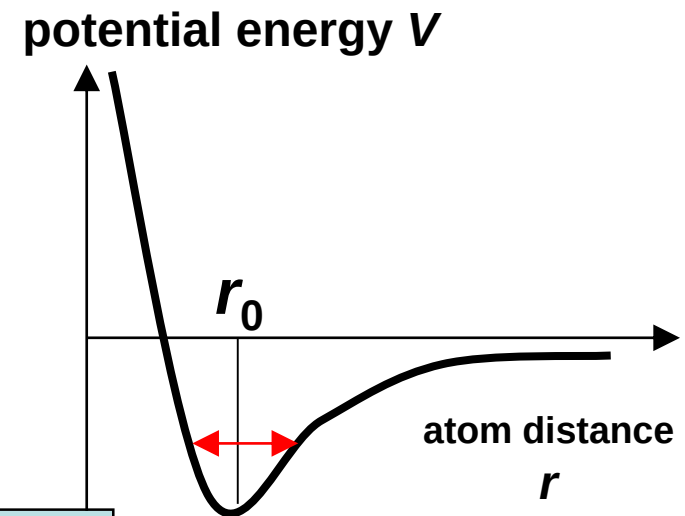
Harmonic Oscillator: Classical Theory

- Vibration amplitude is continuous
- Energy is continuous, and temperature dependent



energy of one spring + one atom
= potential energy + kinetic energy

$$E = \frac{1}{2} Kx^2 + \frac{1}{2} mv^2 = \frac{1}{2} k_B T + \frac{1}{2} k_B T = k_B T$$

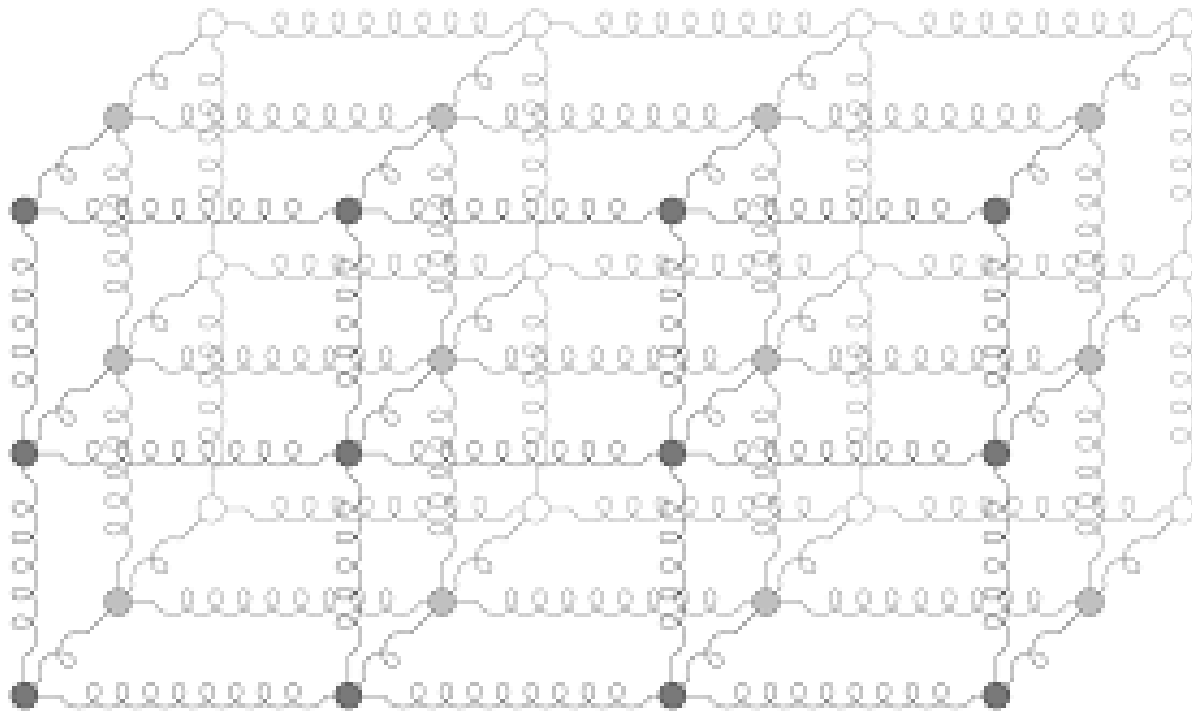


Thermal vibration
around r_0

Energy Equipartition Theorem (能量均分定理)

Internal Energy U 内能

- Total vibration energy of a 3D crystal
 - ▣ all the springs + all the atoms in 3 directions = $3NL$



N - number of primitive cells

L - number of atoms in each primitive cell

Internal Energy U 内能

- Total vibration energy of a 3D crystal

- all the springs + all the atoms in 3 directions = $3NL$

$$U = 3NLk_B T$$

N - # of primitive cells

L - # of atoms in a primitive cell

- Heat capacity (Specific heat) 比热容

- energy per unit of temperature

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V = 3NLk_B$$

Dulong–Petit Law
($3k_B T$ Law)

In the system, every atom contributes an energy of $3k_B T$

Heat Capacity C_V in different units

- **Example: Copper (Cu) at room temperature**
 - ▣ **Dulong–Petit Law ($3k_B$)**

$$C_V = 3k_B = 4.14 \times 10^{-23} \text{ J/K/atom} \quad \text{per atom}$$

$$C_V = 3k_B N_A = 24.9 \text{ J/K/mol} \quad \text{per mole}$$

$$C_V = 24.9 \text{ J/K/mol} \div 0.064 \text{ kg/mol} = 389 \text{ J/K/kg} \quad \text{per unit mass}$$

$$C_V = 389 \text{ J/K/kg} \times 8960 \text{ kg/m}^3 = 3.49 \times 10^6 \text{ J/K/m}^3 \quad \text{per unit volume}$$

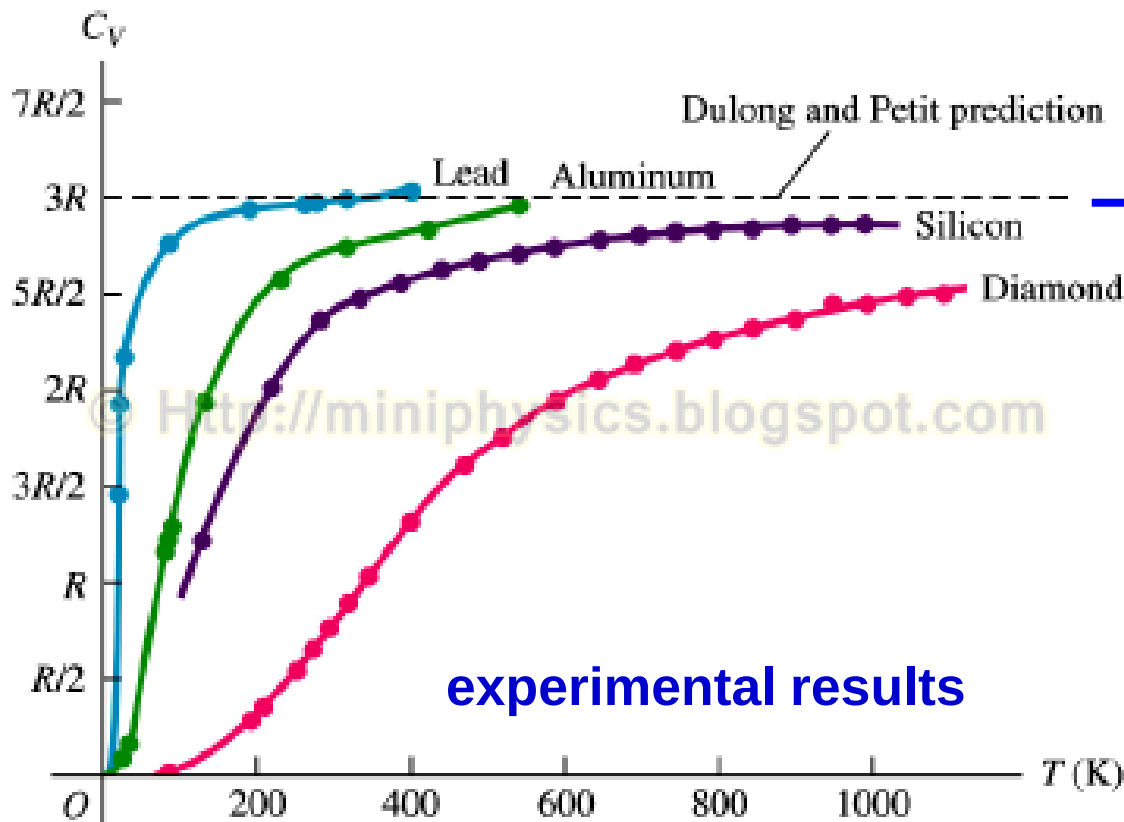
Molar mass of Cu = 64 g/mol

Density of Cu = 8.96 g/cm³

Heat Capacity C_V

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V = 3NLk_B$$

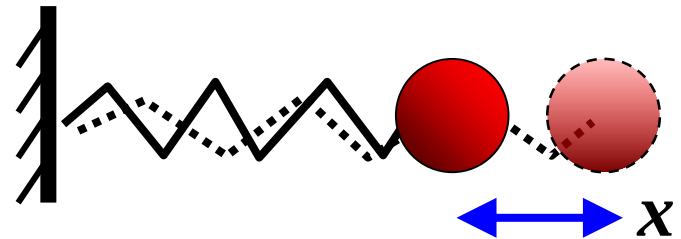
Dulong–Petit Law



The classical Dulong–Petit Law is only valid at high temperatures, but cannot explain the results at low temperatures

Harmonic Oscillator: Quantum Theory

$$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi(x) + V(x) \cdot \psi(x) = E\psi(x)$$

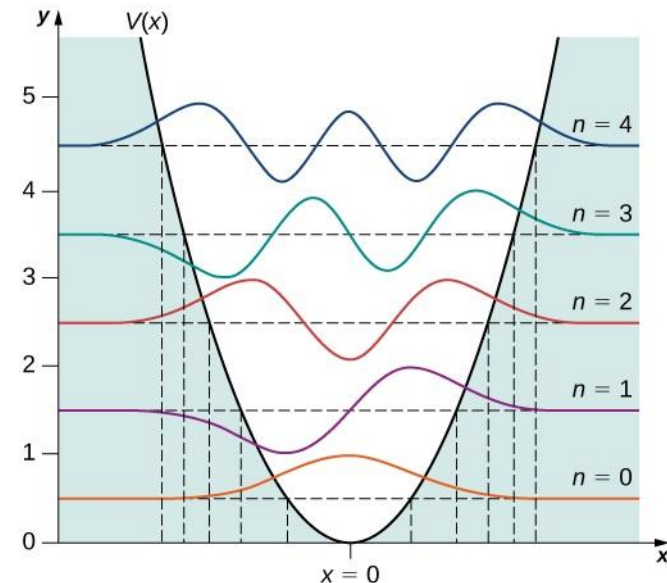


$$V(x) = \frac{1}{2} Kx^2 = \frac{1}{2} m\omega^2 x^2$$

$$\omega = \sqrt{\frac{K}{m}}$$

→ $E_n = \left(\frac{1}{2} + n \right) \hbar\omega \quad n = 0, 1, 2, \dots$

Vibration energy is quantized



Harmonic Oscillator: Quantum Theory

■ Quantum theory

- Vibration energy is quantized
- At each ω state, the energy is the ground state energy ($\hbar\omega/2$) plus energy of n phonons ($\hbar\omega$)

$$E(\omega) = \left(\frac{1}{2} + n \right) \hbar\omega \quad n = 0, 1, 2, \dots$$

- average n in each state follows Bose-Einstein distribution

$$\bar{n} = \frac{1}{e^{\hbar\omega/k_B T} - 1}$$

- If there are N primitive cells, and L atoms in each cell, the total number of states is $3NL$

Internal Energy U 内能

- Internal energy is **the ground state energy plus the energy of all the phonons**

$$U = U_0 + \sum_{i=1}^{3NL} \frac{\hbar \omega_i}{e^{\hbar \omega_i / k_B T} - 1}$$

$$= U_0 + \int_0^{\omega_{\max}} g(\omega) \frac{\hbar \omega}{e^{\hbar \omega / k_B T} - 1} d\omega$$

phonon energy

ground state

quasi-continuous
states

DOS

average phonon number
in each state

heat capacity

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V$$

Heat Capacity C_V

- At high temperature, $T \gg 0 \text{ K}$

$$\begin{aligned}
 U &= U_0 + \sum_{i=1}^{3NL} \frac{\hbar \omega_i}{e^{\hbar \omega_i / k_B T} - 1} \\
 &\approx U_0 + \sum_{i=1}^{3NL} \frac{\hbar \omega_i}{\hbar \omega_i / k_B T} \\
 &= U_0 + 3NLk_B T
 \end{aligned}$$

heat capacity

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V = 3NLk_B$$

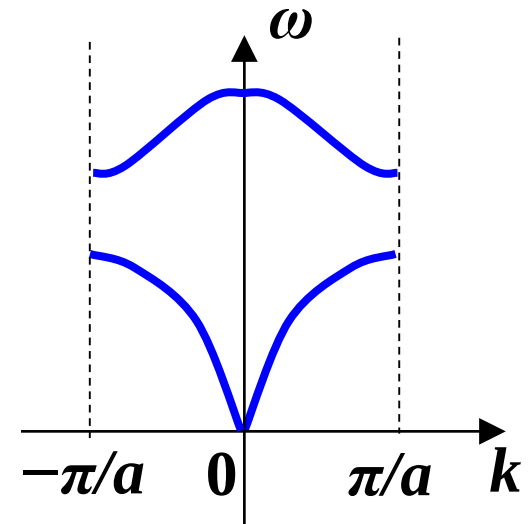
constant

Dulong–Petit Law

Heat Capacity C_V

- At low and medium temperatures

$$\begin{aligned}
 U &= U_0 + \sum_{i=1}^{3NL} \frac{\hbar \omega_i}{e^{\hbar \omega_i / k_B T} - 1} \\
 &= U_0 + \int_0^{\omega_{\max}} g(\omega) \frac{\hbar \omega}{e^{\hbar \omega / k_B T} - 1} d\omega
 \end{aligned}$$



Calculate DOS based
on the phonon diagram

Heat Capacity C_V

- At low and medium temperatures

For optical phonons:

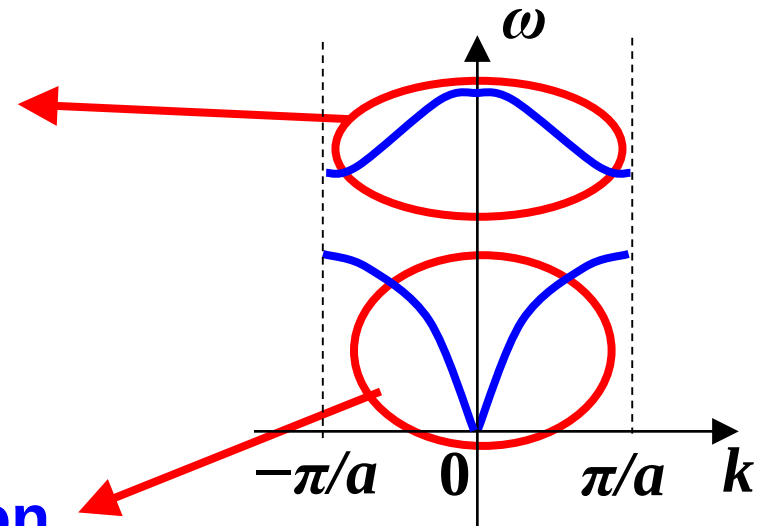
assume all the phonons
have the same ω_0

The Einstein Model
(爱因斯坦模型)

For acoustic phonons:

assume linear ω - k relation

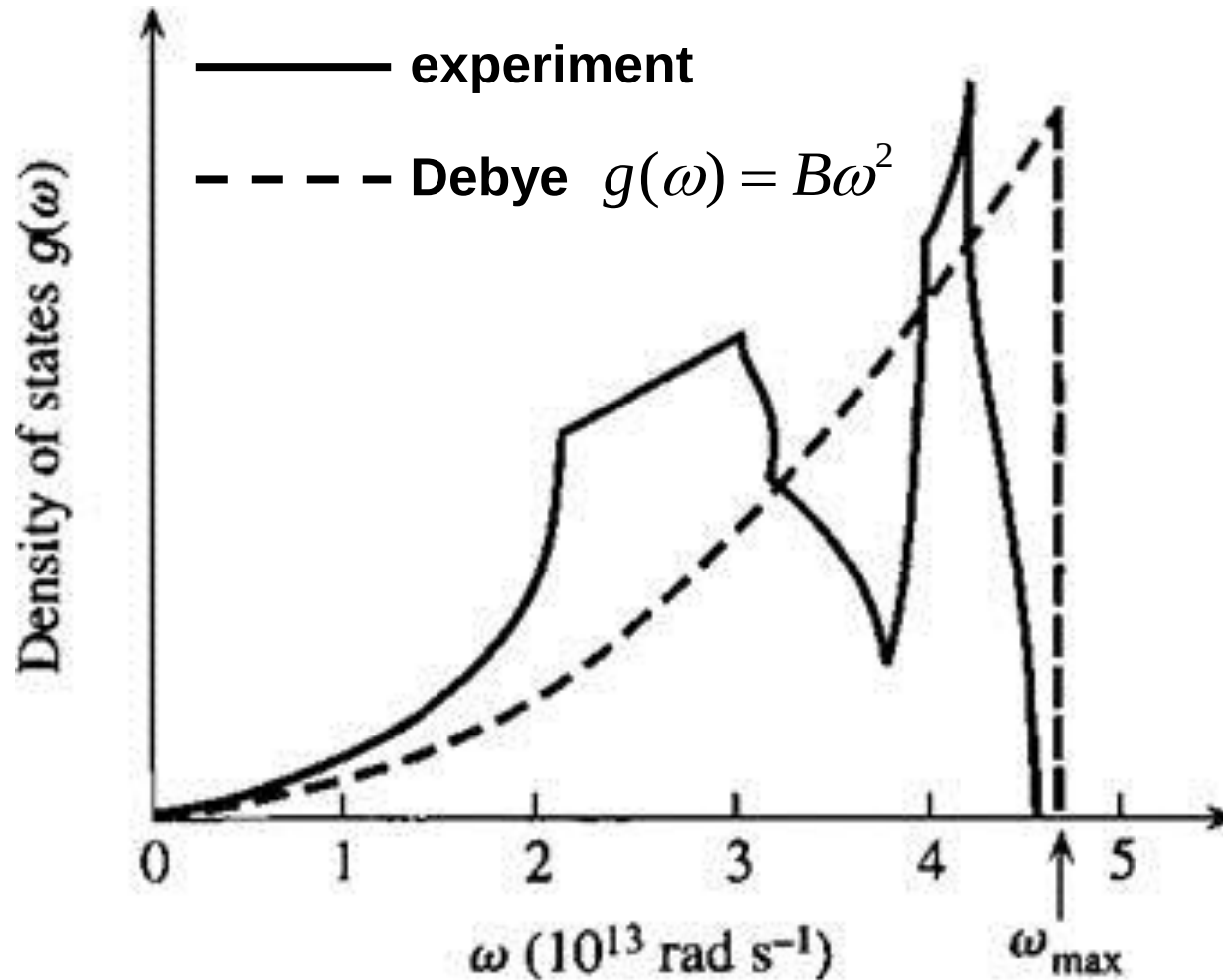
The Debye Model (德拜模型)



$$g(\omega) = B\omega^2$$

Phonon Density of States $g(\omega)$

DOS for copper



Heat Capacity C_V

- **The Debye Model** for the acoustic branch

$$U = U_0 + \int_0^{\omega_{\max}} g(\omega) \frac{\hbar \omega}{e^{\hbar \omega / k_B T} - 1} d\omega$$

$$\approx U_0 + \int_0^{\omega_{\max}} B \omega^2 \frac{\hbar \omega}{e^{\hbar \omega / k_B T} - 1} d\omega$$

At very low temperature, $T \approx 0$ K

heat capacity

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V \approx \frac{12\pi^4}{5} N k_B \left(\frac{T}{\theta_D} \right)^3 \propto T^3$$

**Debye
 T^3 Law**

$$\theta_D = \frac{\hbar \omega_D}{k_B} = \frac{\hbar v_g}{k_B} \left(\frac{6\pi^2 N}{V} \right)^{1/3}$$

Debye Temperature

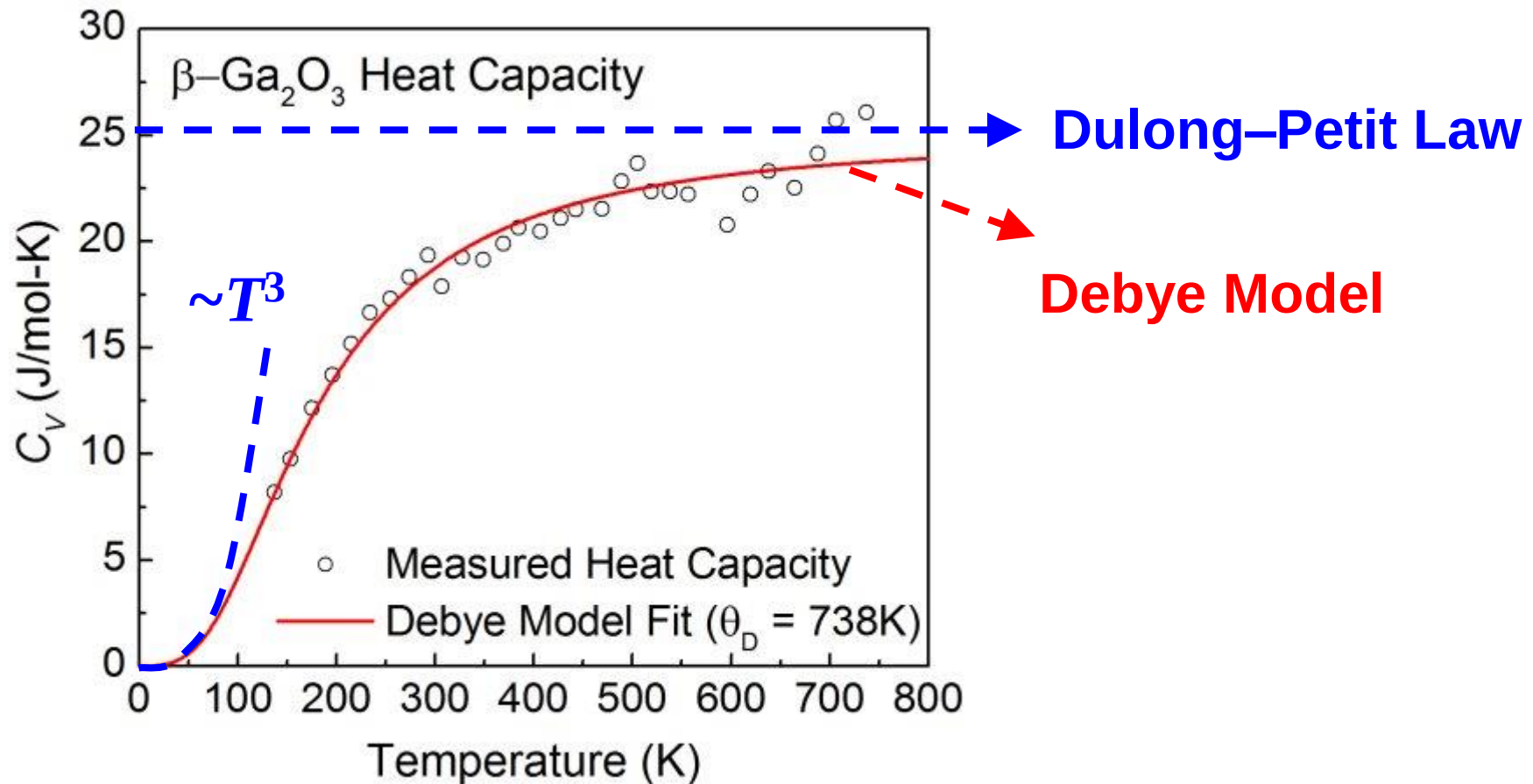
Heat Capacity C_V

- **The Debye Model** for the acoustic branch
- **Debye Temperature** is around room temperature for most materials

$$\theta_D = \frac{\hbar \omega_D}{k_B} = \frac{\hbar v_g}{k_B} \left(\frac{6\pi^2 N}{V} \right)^{1/3}$$

	θ_D (K)
C (diamond)	2230
Si	645
Al	428
Cu	343

Heat Capacity C_V - Example



Heat Capacity C_V

- **The Einstein Model** assumes all the phonons in an optical branch have a constant frequency ω_0

$$U = U_0 + \sum_{i=1}^N \frac{\hbar \omega_i}{e^{\hbar \omega_i / k_B T} - 1} = U_0 + N \frac{\hbar \omega_0}{e^{\hbar \omega_0 / k_B T} - 1}$$

heat capacity
(for one branch)

$$\begin{aligned} C_V &= \left(\frac{\partial U}{\partial T} \right)_V = N k_B \left(\frac{\hbar \omega_0}{k_B T} \right)^2 \frac{e^{\hbar \omega_0 / k_B T}}{(e^{\hbar \omega_0 / k_B T} - 1)^2} \\ &= N k_B \left(\frac{\theta_E}{T} \right)^2 \frac{e^{\theta_E / T}}{(e^{\theta_E / T} - 1)^2} \end{aligned}$$

$$\theta_E = \frac{\hbar \omega_0}{k_B}$$

Einstein Temperature

Heat Capacity C_V

- Heat capacity is the combination of all the acoustical and optical phonons

$$C_V = C_{V,acoustic} + C_{V,optical}$$

For optical phonons:

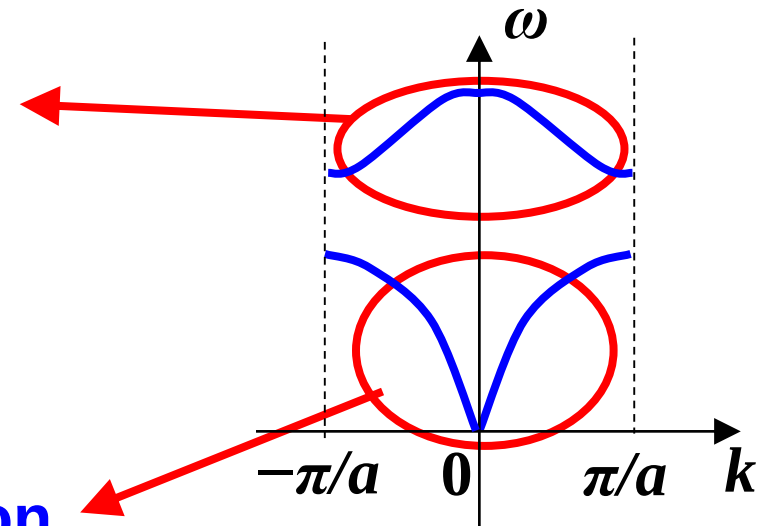
assume all the phonons
have the same ω_0

The Einstein Model
(爱因斯坦模型)

For acoustic phonons:

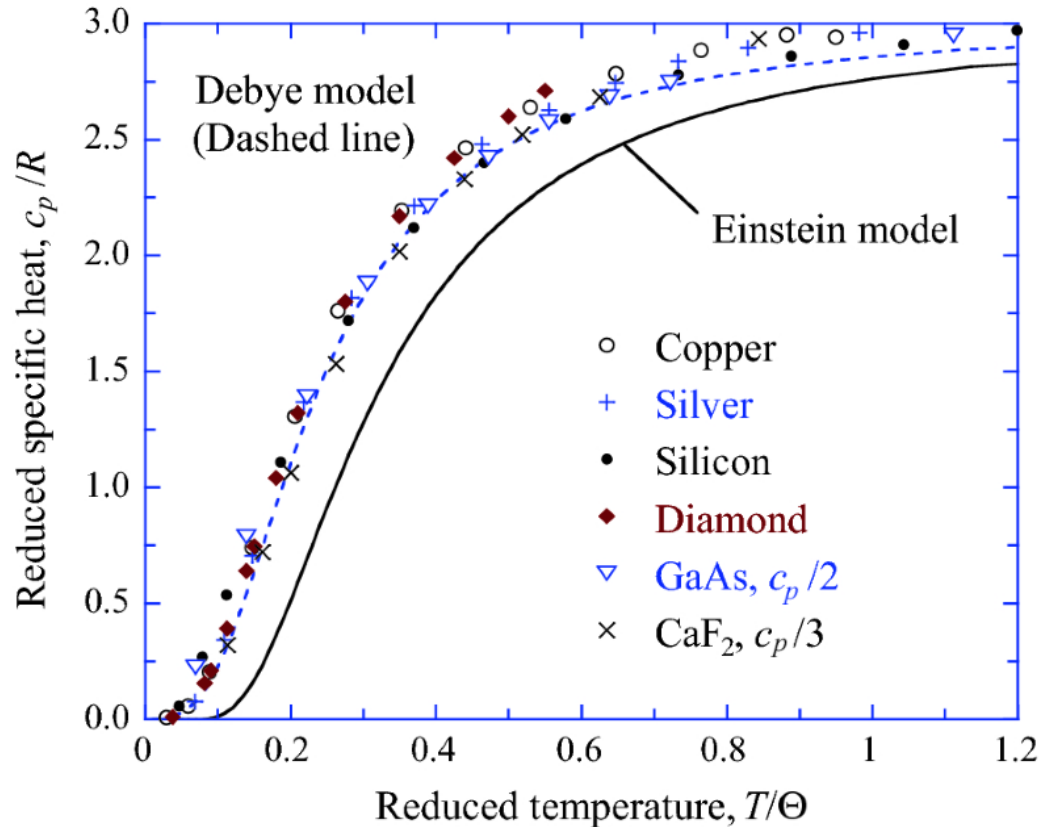
assume linear ω - k relation

The Debye Model (德拜模型)



$$g(\omega) = B\omega^2$$

Heat Capacity C_V - Example



**The Debye Model
matches better
with experimental
results**

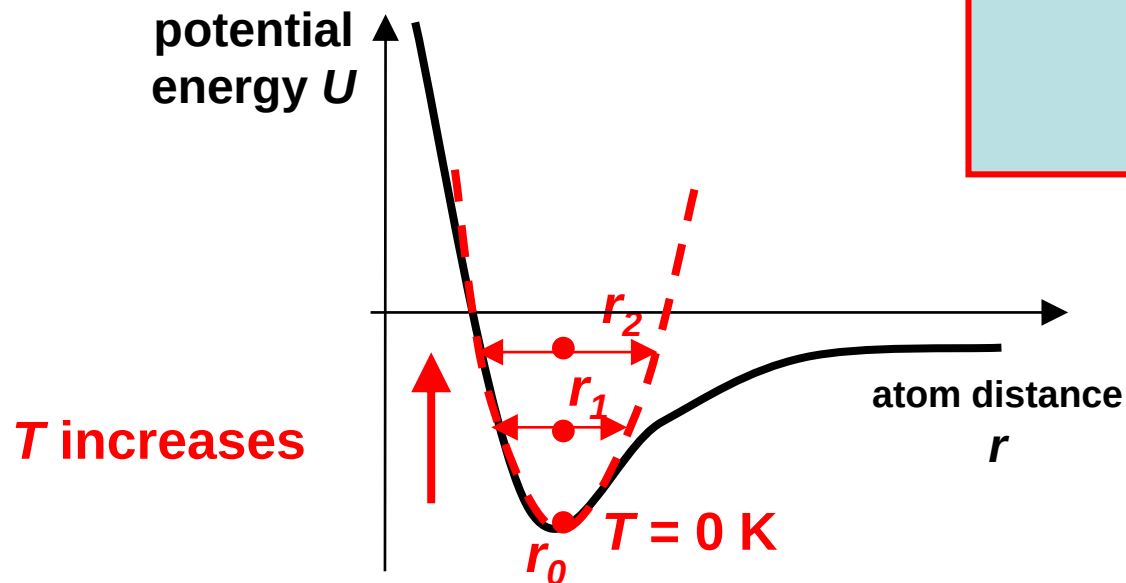
Thermal Properties

- Heat Capacity (Thermal Capacity) 热容
- Thermal Expansion 热膨胀
- Thermal Conductivity 热导
- ...

Thermal Expansion 热膨胀

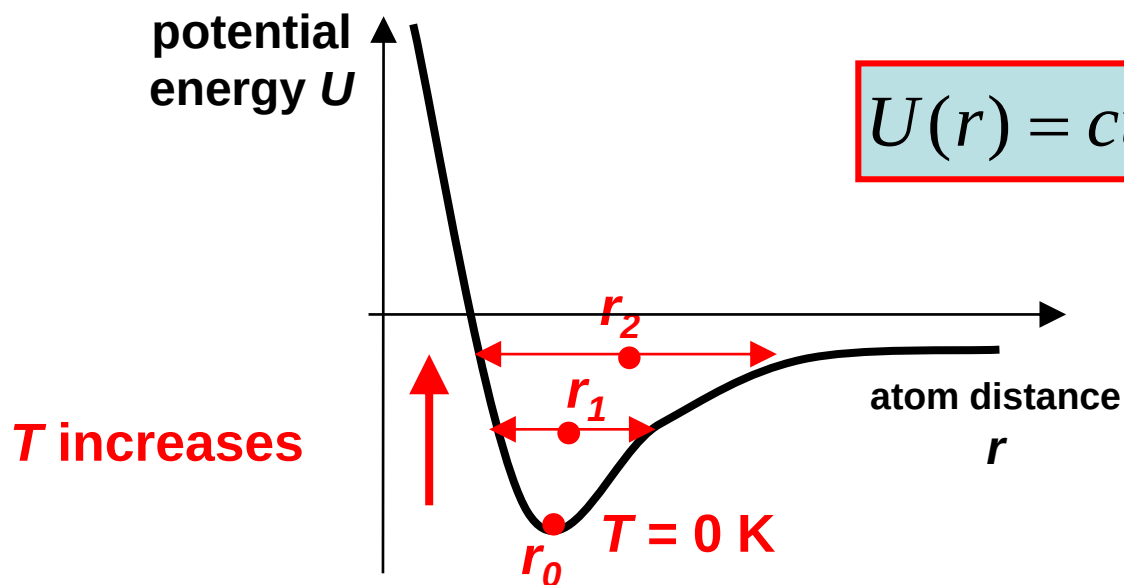
- Thermal expansion cannot be explained by the **harmonic approximation**

$$U(r) = U_0 + \frac{1}{2} K (r - r_0)^2$$
$$= U_0 + \frac{1}{2} K \cdot u^2$$



Thermal Expansion 热膨胀

- Thermal expansion cannot be explained by the **harmonic approximation**
- Thermal expansion originates from the **anharmonic** nature of the potential
- Vibration increases with temperature



$$U(r) = cu^2 - gu^3 - fu^4 - \dots$$

Thermal Properties

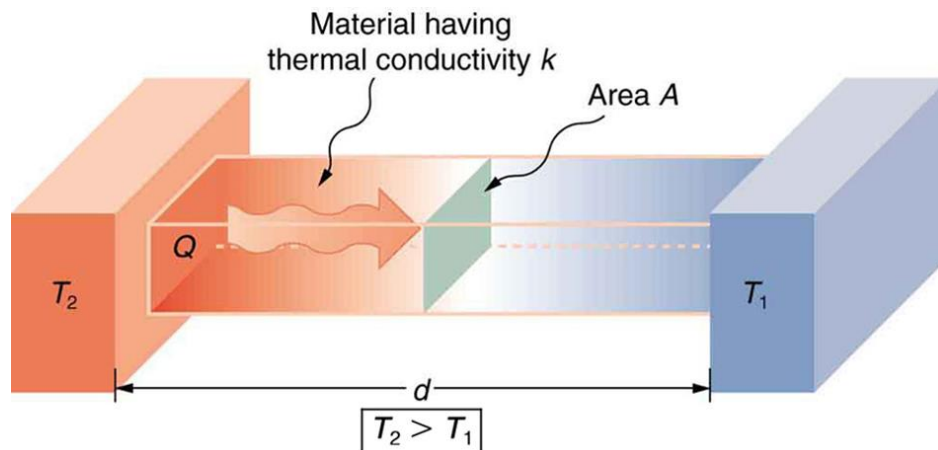
- Heat Capacity (Thermal Capacity) 热容
- Thermal Expansion 热膨胀
- Thermal Conductivity 热导
- ...

Thermal Conduction 热导

■ Fourier's Law

- heat flux is proportional to the temperature gradient

$$Q = -\kappa \cdot \frac{dT}{dx}$$



<https://www.khanacademy.org>

Q - heat flux (W/m^2)
 κ - thermal conductivity (W/m/K)
 T - temperature (K)

Thermal Conductivity 热导率

- Thermal conductivity κ

$$\kappa = \frac{1}{3} C_V v_g l = \frac{1}{3} C_V v_g^2 \tau_p$$

$$l = v_g \tau_p$$

Ashcroft & Mermin, p20

C_V - thermal capacity

v_g - sound speed

l - phonon mean free path

τ_p - phonon relaxation time

l and τ_p is dependent on crystal structure, defects, impurities, ...

Q: Which material has the highest thermal conductivity?

Thermal Properties

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- For insulators, there are no free electron. Thermal properties of **lattice vibration (phonons)** dominate.
- For metals,
thermal properties = **phonon part + free electron part**

Thermal capacity

$$C_V = C_{V,p} + C_{V,e}$$

Thermal conductivity

$$\kappa = \kappa_p + \kappa_e$$

Fundamentals of Solid State Physics

Thermal Properties of Free Electrons

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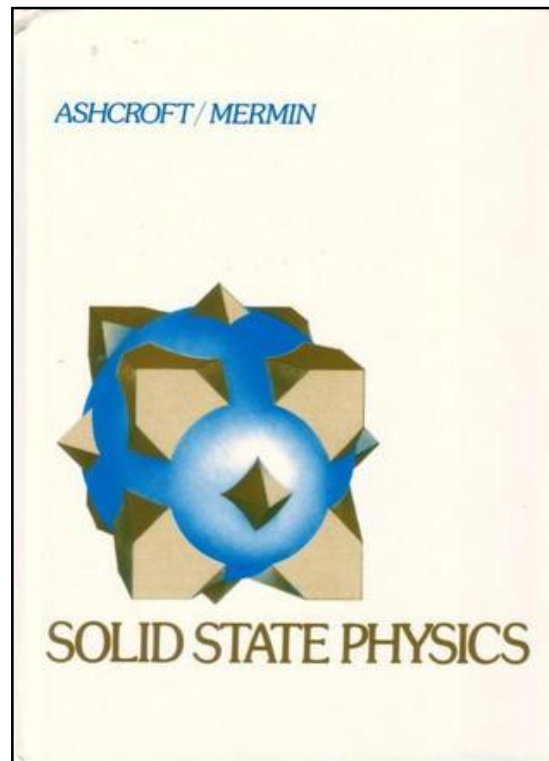
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Review

- Lecture 3.1, Sommerfeld Model
- Ashcroft & Mermin, Chapter 2



Density of States (DOS) 态密度

$$g(E) = \frac{dn}{dE}$$

DOS - number of energy states/levels per unit energy in $[E, E+dE]$, per unit volume

$$k = (3\pi^2 n)^{1/3}$$

$$E = \frac{\hbar^2 k^2}{2m_e}$$



$$n = \frac{1}{3\pi^2} \left(\frac{2m_e}{\hbar^2} \right)^{3/2} E^{3/2}$$

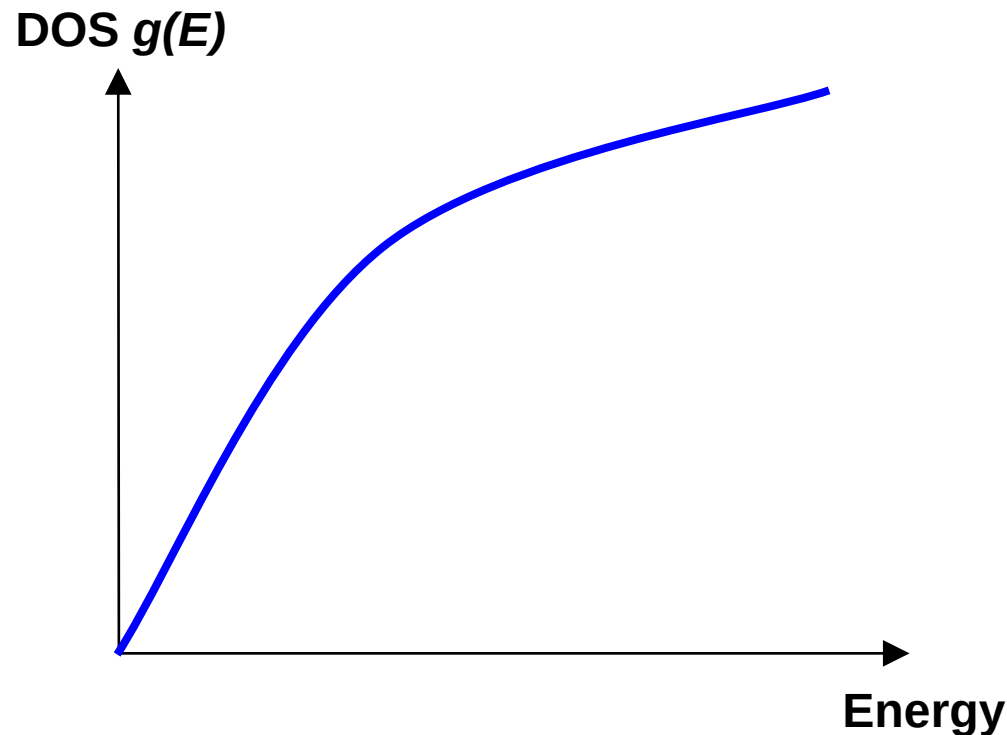
n - free electron density



$$g(E) = \frac{dn}{dE} = \frac{1}{2\pi^2} \left(\frac{2m_e}{\hbar^2} \right)^{3/2} E^{1/2}$$

Density of States (DOS) 态密度

$$g(E) = \frac{dn}{dE} = \frac{1}{2\pi^2} \left(\frac{2m_e}{\hbar^2} \right)^{3/2} E^{1/2}$$



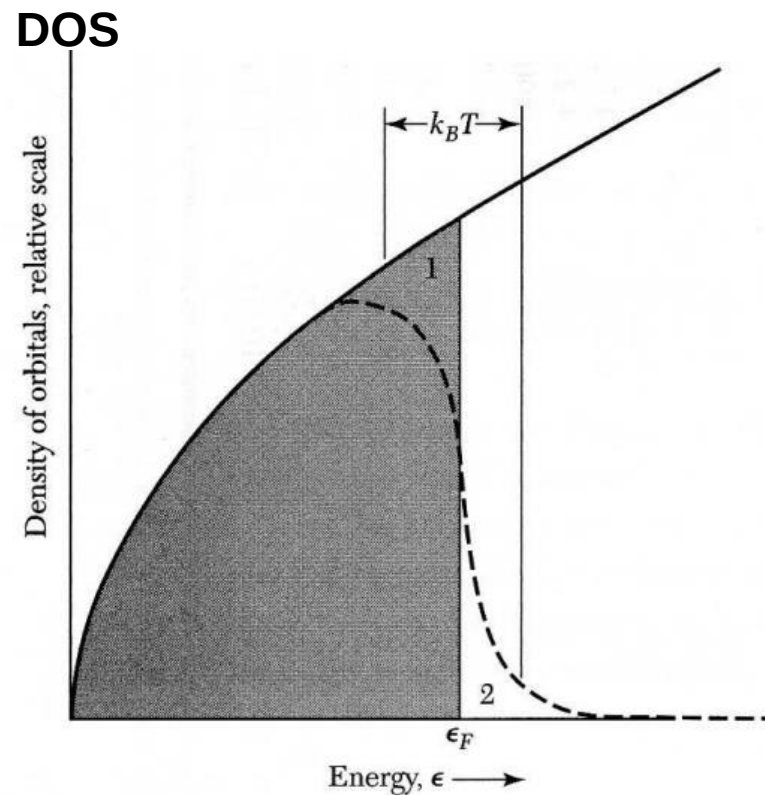
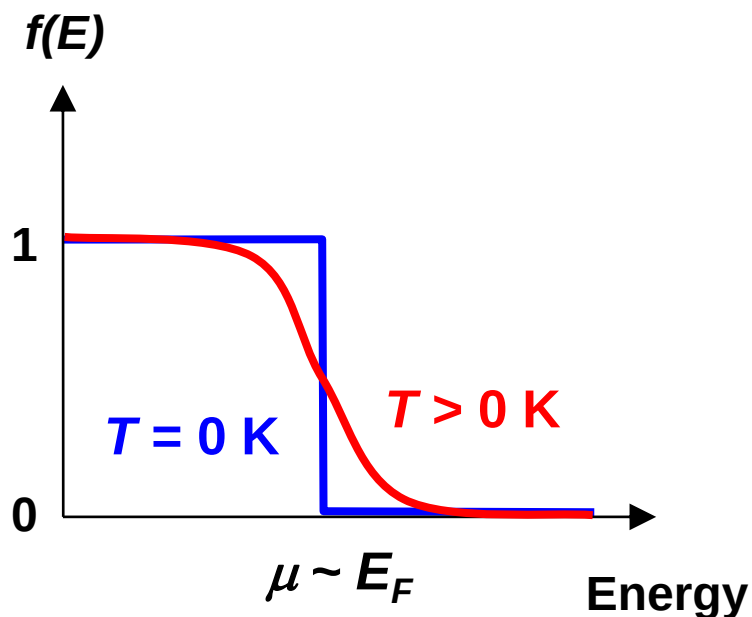
3D Free Electrons

Density of Electrons

Density of electrons = DOS * probability

$$f(E)g(E)$$

$$f(E) = \frac{1}{e^{(E-\mu)/k_B T} + 1}$$



When $T > 0 \text{ K}$, some electrons are excited to higher states (from 1 to 2)

Internal Energy 内能

- Internal energy is **the energy of all the free electrons**

The diagram illustrates the components of the internal energy equation $U = \int_0^{+\infty} g(E) f(E) E dE$. Three blue arrows point from the equation to its constituent parts:

- An arrow from $g(E)$ points to the text **DOS** (Density of States).
- An arrow from $f(E)$ points to the text **average electron number in each state (Fermi-Dirac Distribution)**.
- An arrow from E points to the text **electron energy**.

The Sommerfeld Model

Internal Energy 内能

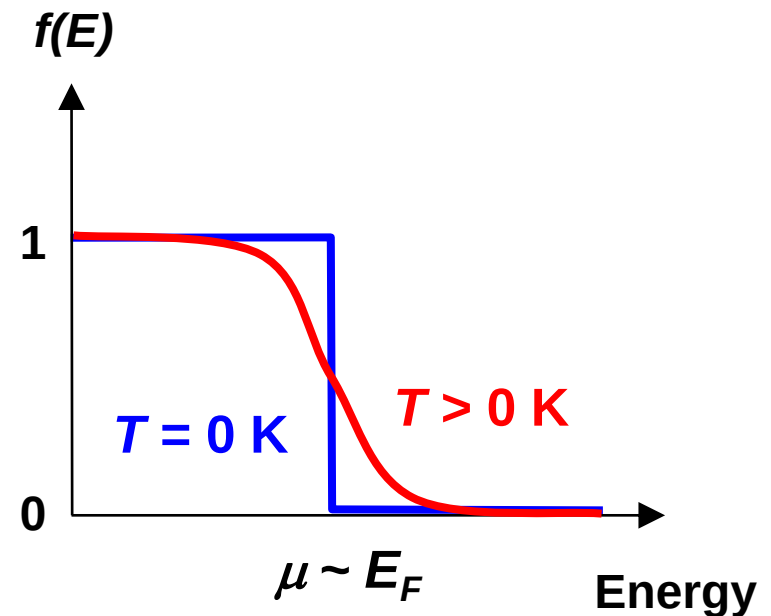
- When $T = 0$ K

$$U_0 = \int_0^{E_F} g(E) \cdot E dE$$

$$= \frac{3}{5} E_F$$

Homework 4.4

$$f(E) = \frac{1}{e^{(E-\mu)/k_B T} + 1}$$



Heat Capacity C_V

- When $T > 0$ K

Ashcroft & Mermin, p46

$$U = U_0 + \frac{\pi^2}{6} (k_B T)^2 g(E_F)$$

heat capacity

$$C_{V,e} = \left(\frac{\partial U}{\partial T} \right)_V = \frac{\pi^2}{2} \frac{T}{T_F} n k_B \propto T$$

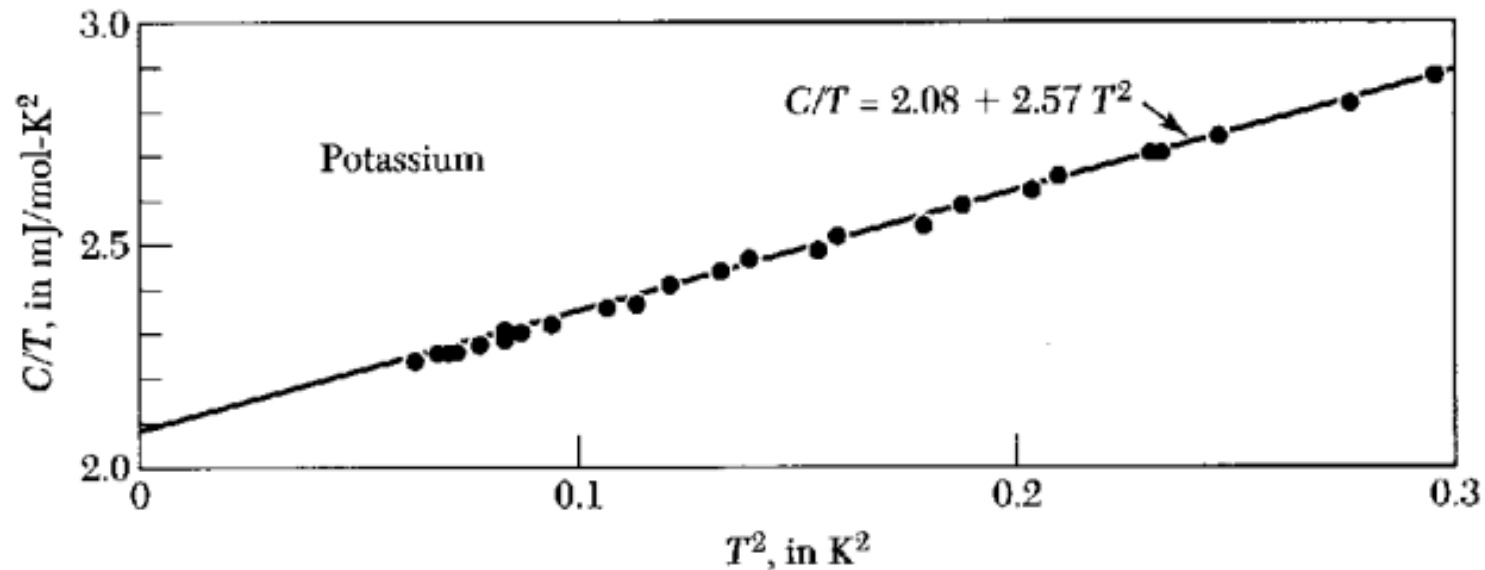
T_F - Fermi temperature ($\sim 10^4$ K)

Only a few electrons around E_F contribute to $C_{V,e}$.
 At room temperature, for free electrons $C_{V,e} \ll Nk_B$
 much smaller than C_V from phonons

Heat Capacity C_V

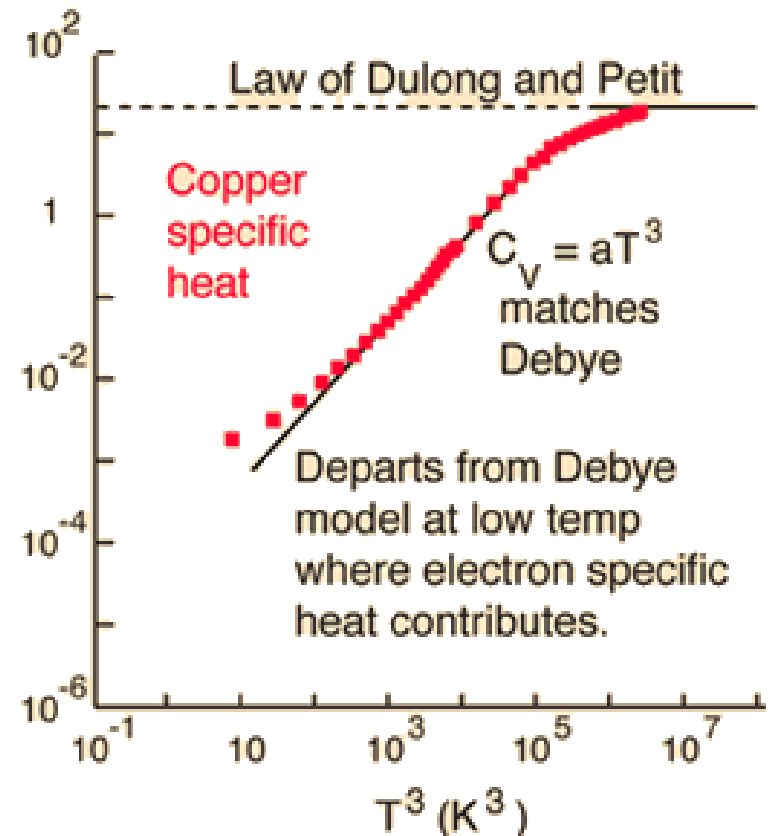
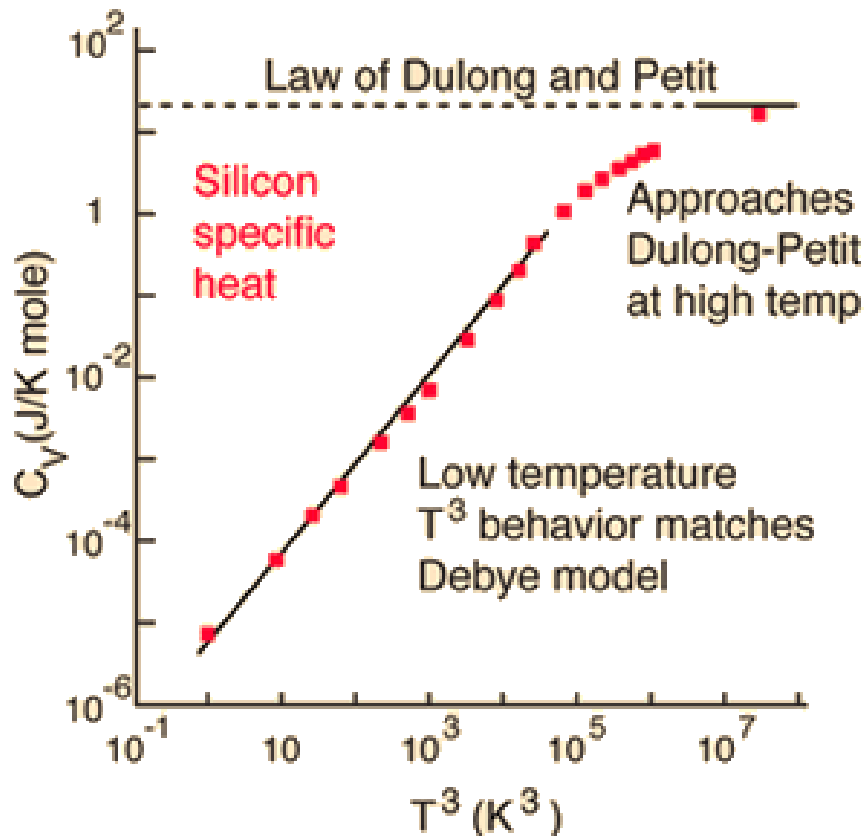
- For metals at very low temperature $T \sim 0$ K
 - Thermal properties = phonon part + free electron part

$$C_V = C_{V,p} + C_{V,e} = AT^3 + \gamma T \quad \rightarrow \quad C_V / T = AT^2 + \gamma$$



Heat Capacity C_V

- More examples
 - silicon vs. copper



Thermal Conductivity 热导率

- Thermal conductivity κ for free electrons

$$\kappa_e = \frac{1}{3} C_{V,e} v_F l = \frac{1}{3} C_{V,e} v_F^2 \tau_e$$

$$l = v_F \tau_e$$

Ashcroft & Mermin, p20

C_V - thermal capacity

v_F - Fermi velocity

l - electron mean free path

τ_e - electron relaxation time

Thermal Conductivity 热导率

- Thermal conductivity κ for metals

$$\kappa = \kappa_p + \kappa_e = \frac{1}{3} C_{V,p} v_g^2 \tau_p + \frac{1}{3} C_{V,e} v_F^2 \tau_e$$

For conductive metals like Cu or Ag, $v_F \gg v_g$
electron part dominates

Thermal Conductivity 热导率

- Ratio of thermal conductivity κ and electron conductivity σ for certain metals is a constant
 - Lorentz number L

$$L = \frac{\kappa}{\sigma T}$$

Wiedemann and Franz law in 1853

metals at 300 K	Cu	Ag	Au	Al	Fe	Pb
L ($10^{-8} \text{ W} \cdot \Omega / \text{K}^2$)	2.30	2.31	2.35	2.23	2.47	2.45

Q: Which metal has the highest thermal conductivity?

Thermal Conductivity 热导率

- Relationship of thermal conductivity κ and electron conductivity σ for certain metals
 - Lorentz number L

metals at 300 K	Cu	Ag	Au	Al	Fe	Pb
L ($10^{-8} \text{ W} \cdot \Omega / \text{K}^2$)	2.30	2.31	2.35	2.23	2.47	2.45

$$L = \frac{\kappa_e}{\sigma T} = \frac{\pi^2}{3} \left(\frac{k_B}{e} \right)^2 = 2.44 \times 10^{-8} \text{ W} \cdot \Omega / \text{K}^2$$

Homework 8.7

Not correct for most other materials, because only the free electron part satisfies this.

Thermal Conductivity 热导率

	κ (W/m/K)
C (diamond)	2000
Cu	400
C (graphite)	~200
Si	130
glass	1
paper	0.05

Q: High thermal conductivities of diamond and graphite have different origins. Why?

Summary

- Thermal properties are the combinations of properties of **lattice vibration (phonons)** and **free electrons**
- For insulators, there are no free electron. Thermal properties of **lattice vibration (phonons)** dominate.
- For metals,
thermal properties = **phonon part + free electron part**

Thermal capacity

$$C_V = C_{V,p} + C_{V,e}$$

Thermal conductivity

$$\kappa = \kappa_p + \kappa_e$$

Thank you for your attention